

Complexifying an Optical Neural Network Based on Synthetic Nonlinear Photonic Lattices

**A Report Outlining Work Done Based on the Original Paper and Thoughts Going
Forward**

By

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Abstract

This report intends on extending from the works of Artem V. Pankov et al., where a synthetic photonic lattice (SPL) is used to generate a feed-forward optical neural network (ONN) through time-domain multiplexing [1].

This report briefly summarizes the work I've completed over the summer of 2023 with the INRS, including a summary of the original setup, its connection to an ONN and the code I've made to replicate the original SPL. An attempt at complexifying the original setup is outlined in this paper by using multiple pairs of loops and feeding the pulses from one setup to the next, where coefficients are of different values.

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How Does the Original Setup Work?

There are two loops coupled together by a coupling coefficient Θ , where one loop is longer than the other. To begin, a pulse is sent into the longer loop, denoted as pulse v . As the pulse travels through the loop in a clockwise direction, the pulse is sent through a non-linear element f and then is split into two weaker pulses via the coupler Θ . The weaker pulse in the longer loop continues its journey to the linear phase Ψ and then completes a loop. Additionally, the weaker pulse in the shorter loop, denoted as u , goes through a linear phase ϕ and finally a gain parameter γ to modulate the energy of each pulse to complete a single loop.

This process is repeated many times until a pulse train is created. We note that due to the different lengths of each loop, the pulses obtained from the original pulse splitting will not couple again. The pulses u and v will continue to split creating more and more pulses until a pulse train is formed.

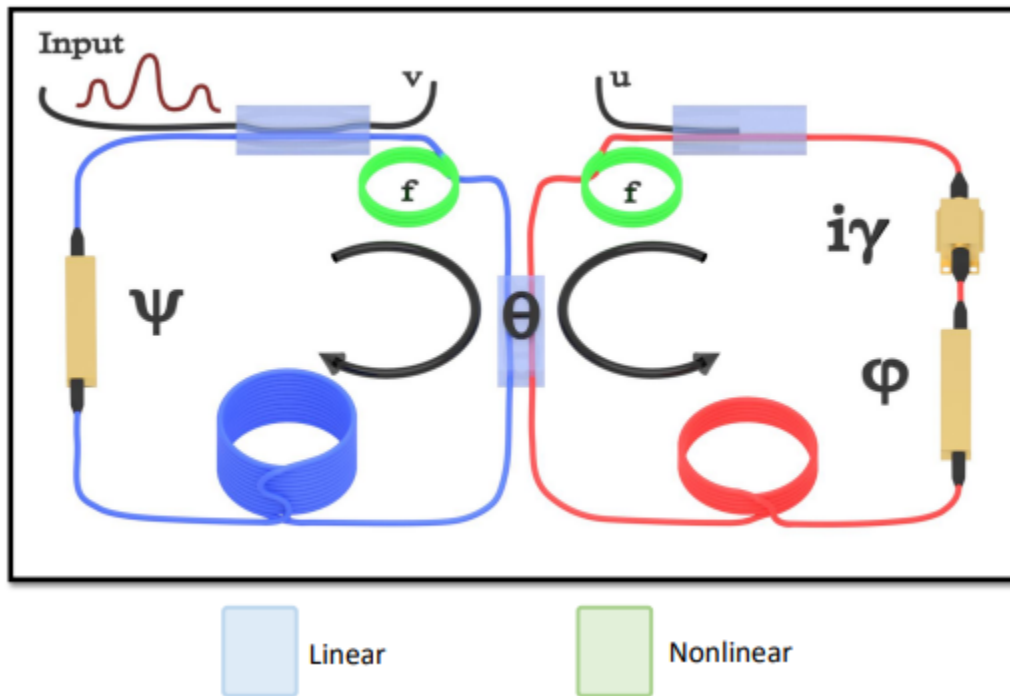


Figure 1: The original setup in “Optical Neural Network Based on Synthetic Nonlinear Photonic Lattices” outlining the placement of various parameters [1].

How this forms an ONN

This synthetic photonic lattice generates a feed-forward deep neural network through time-domain multiplexing. We denote an arbitrary layer in the network as m , which corresponds with the m^{th} roundtrip of the pulse, as well as each neuron in the layer as n , which corresponds to the n^{th} pulse in the pulse train.

The interaction between the short and long pulses is as follows: The n^{th} long pulse v will couple with the $n-1$ short pulse u due to the difference in lengths between loops; When the n^{th} long pulse v arrives back at the coupler θ , enough time has passed such that the n^{th} short pulse u has already passed the θ coupler in the same round trip as the aforementioned v , thus the long pulse v has to interact with the pulse behind the n^{th} short pulse u , which is defined to be the $n-1$ short pulse u . Ultimately, a roundtrip through the shorter loop increases m by one and decreases n by one, whereas a roundtrip through the longer loop increases m by one and increases n by one, as shown in Figure 2 below. The superposition of pulses between loops can be calculated with the equations below:

$$\begin{aligned} u_n^{m+1} &= \exp(i\phi_n^m + \gamma_n^m) [f(|u_{n-1}^m|^2) u_{n-1}^m \cos(\theta_{n-1}^m) + \\ &\quad i f(|v_{n-1}^m|^2) v_{n-1}^m \sin(\theta_{n-1}^m)] , \\ v_n^{m+1} &= \exp(i\psi_n^m) [f(|v_{n+1}^m|^2) v_{n+1}^m \cos(\theta_{n+1}^m) + \\ &\quad i f(|u_{n+1}^m|^2) u_{n+1}^m \sin(\theta_{n+1}^m)] , \end{aligned}$$

Where $f(|x|^2) = \exp(i\alpha|x|^2)$, where α is a nonlinear coefficient. We also note the paper uses the following notation:

For a complex pulse amplitude of short pulse u , the n^{th} complex pulse amplitude on its m^{th} roundtrip is written as such: u_n^m (as seen above)

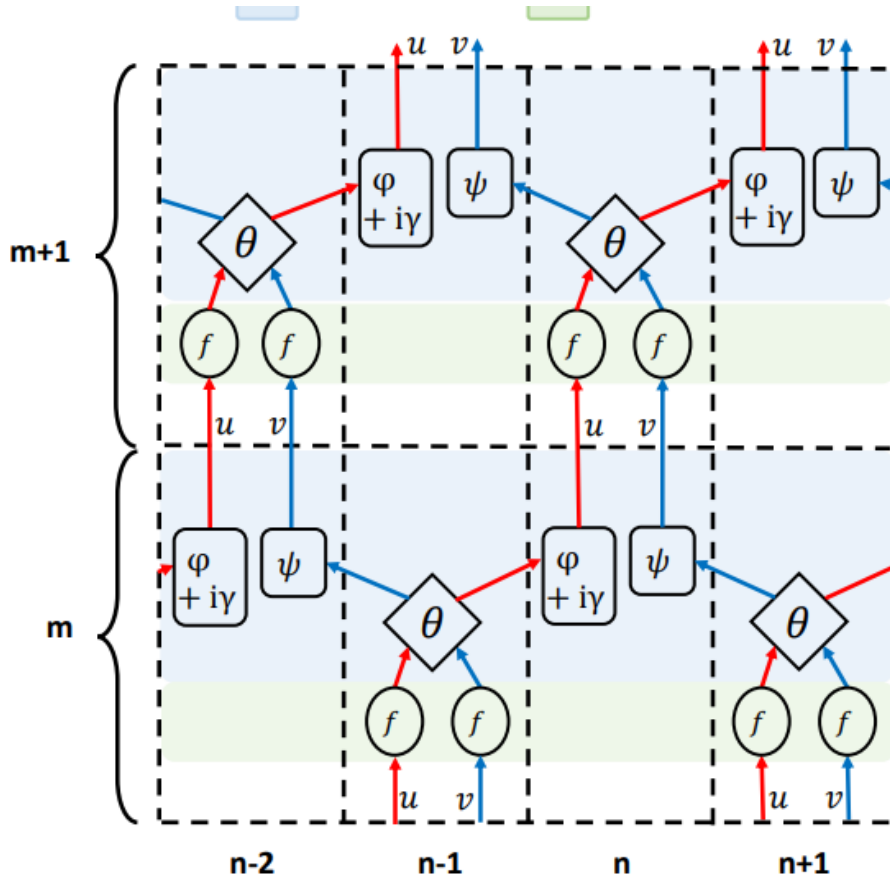


Figure 2: The ONN formed by the SPL, where u is the pulse in the short loop and v is the pulse in the long loop. A layer is completed once a roundtrip has been completed by the pulses [1].

Going Beyond the Source Material

The main goal of improving upon the original paper was to add more loops to the original lattice and look into how the apparatus changes/improves. After performing further research into the addition of loops to the original 2 loop system, it was concluded that following the original setup it would prove to be difficult to successfully implement for the following reasons:

1. The coupling of pulses would have to happen at the same time for both couplers in order to make a transparent divide between layers. This could be done by making the third loop the same length as the long loop v , such that the setup from left to right goes: long loop, short loop then long loop again.
2. With 3 loops in a line, pulse amplitudes from each loop do not all superimpose at once, rather 2 pulses experience superposition at a time. This causes some pulses to travel a short distance in the middle loop to travel a much shorter distance to go from one coupler to another. To clarify, in the far left and far right loops, each pulse would complete an entire roundtrip in their respective loops, whereas a pulse in the centre would travel only half of the loop due to the couplers intersecting the middle loop twice, requiring a gain modulator in the middle loop.
3. In a triangular setup, where all loops intersect through one coupler, the direction of flow for each loop would not flow in the proper manner, as shown in Figure 4 below. Based on the original setup, you could not have all 3 rings agree with the direction of flow.

After thorough research of various papers, it was decided to add multiple copies of the original pair of loops such that one pair of loops completes its M roundtrips and then have the pulse trains from each loop fed into another pair of loops with different coefficients. This could potentially lead to the increase in dimensions of the ONN using multiple pairs of the loop connected to one another, as shown in Figure 3 below:

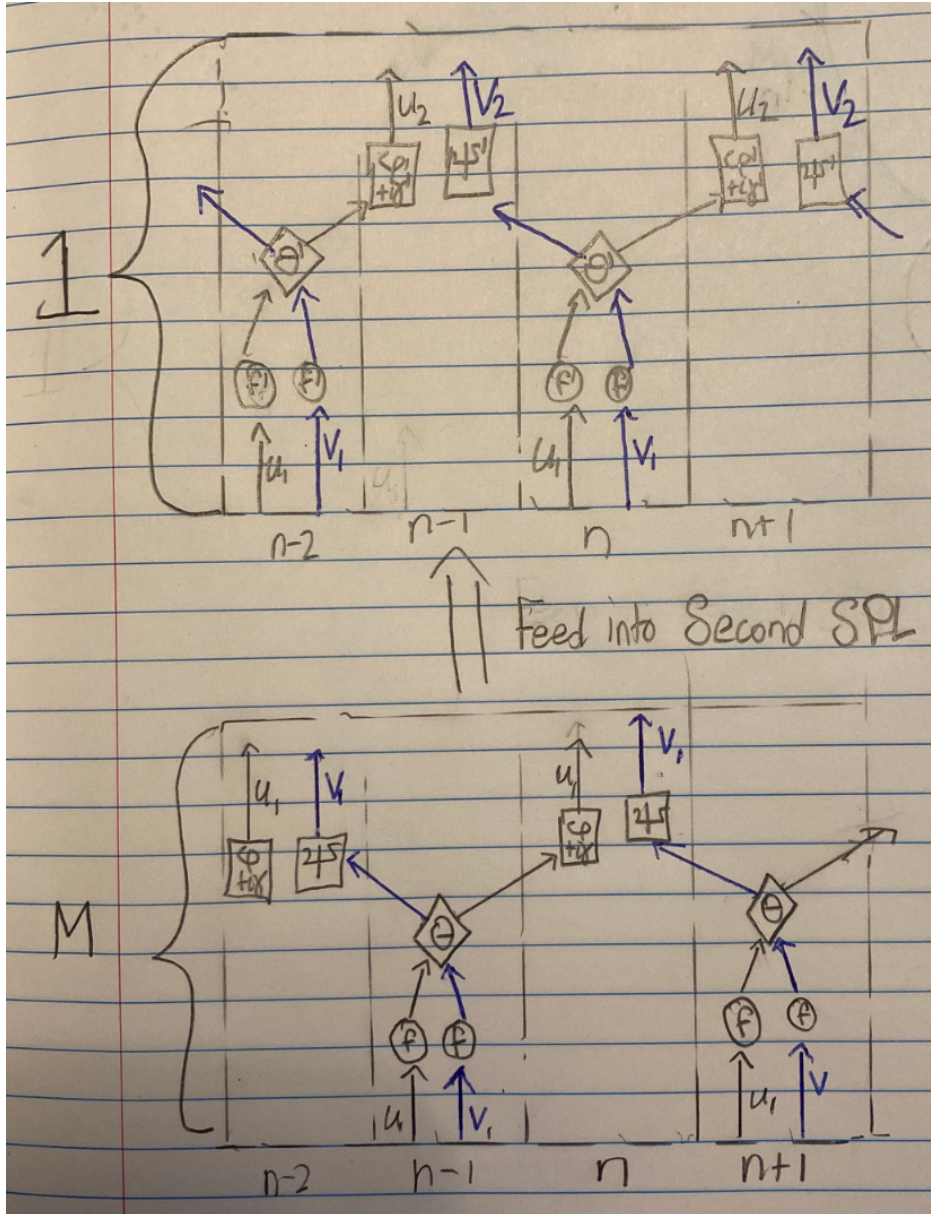


Figure 3: Image outlining how the pulses from a previous would be implemented into the next lattice with different parameters

Another paper was found with a similar premise, where instead of having originally 2 different paths of different lengths, the paper contained 3 paths each of varying length, referred to as a single resonator network with two delay lines [2]. This leads to a similar neural network with an extra pulse amplitude for the superposition of amplitudes, complexifying the system.

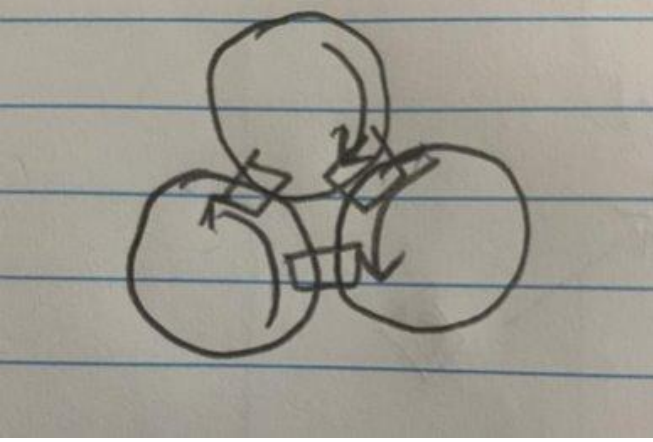


Figure 4: The bottom left ring flows properly with the top ring however it does not flow correctly with the right ring

Explaining the Code

Using the formulas for the superpositions of the complex pulse amplitudes, I've created a program to emulate the original synthetic photonic lattice, which can be modified for any given amount of roundtrips, M , and number of pulses in each loop, N . Following the original paper's example, I've set $M=43$ and $N=21$ and randomly generate complex pulse amplitudes with similar ranges to the example shown in Section III A. "Signal restoration in telecommunication fiber links". For the matrices involving the complex pulse amplitudes, a column represents a roundtrip and a row represents a specific pulse.

I've also created a dictionary in the code to make it easier to access any necessary pulse in any roundtrip as well as feed the last column of both complex pulse amplitudes, u_n^M , v_n^M , into the next pair of loops. That way, the next pair of loops will start with the previous pair of loops' amplitudes after M roundtrips, where the values of the linear phases and gain are different from the previous pair of loops.

There are some current issues with the code however:

- Complex pulse amplitudes continue to grow in magnitude with each roundtrip, which is a problem. Potential reasons for this include that the value of the coefficients may not be a suitable range, as well as the possibility that the superposition of the amplitudes does not factor in the loss in overall intensity pulses have after each roundtrip. This issue has temporarily been resolved by normalizing each pulse amplitude after a roundtrip, however it is not clear how this will impact the ONN.
- The training and learning for the ONN needs to be implemented. The specifics of the kind of learning and training the ONN has is outlined in the paper. Due to time constraints and unfamiliarity, I was not able to accomplish this.
- The values of the modulators as well as the coupler are ambiguous outside of boundary conditions. If this code is looked into deeper, I suggest replacing the placement values I have for each coefficient.

Conclusion & Final Remarks

The setup suggested above for complexifying the network could potentially help with expanding the dimensionality of the ONN by using various pairs of loops connected to one another. This report focuses on the logic used for connecting just one loop to another, which is essential for expanding this system.

With regards to the physical implementation of the system, one possibility for having 2 pairs of loops connected to one another could be done with 1 pair of loops; After the desirable amount of roundtrips, one could potentially set a delay for the pulse trains in each loop and manipulate the parameters of the SPL such that it resembles the desired second system. That way, once the pulse train continues through the lattice, it's as if it's in a new pair of loops.

Furthermore, I've provided a folder of numerous papers that were useful for the research conducted on this project. If this project were looked into further, I recommend browsing the aforementioned papers.

References

- [1] Pankov, Artem V., et al. "Optical Neural Network Based on Synthetic Nonlinear Photonic Lattices." *Physical Review Applied*, vol. 17, no. 2, 2022, <https://doi.org/10.1103/physrevapplied.17.024011>.
- [2] Peng, Bo, et al. "Optical Neural Network Architecture for Deep Learning with Temporal Synthetic Dimension." *Chinese Physics Letters*, vol. 40, no. 3, 2023, p. 034201, <https://doi.org/10.1088/0256-307x/40/3/034201>.